2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)
16 – 18 April 2026, Chittagong, Bangladesh

Human–AI Interaction in Healthcare: A User-Centered Evaluation of Trust and Usability in Explainable AI Interfaces

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***Abstract*—Artificial intelligence (AI) technology has been applied more and more in healthcare applications for decision support, the lack of transparency behind many AI systems often results in diminished trust and suboptimal adoption. In the human–computer interaction (HCI) domain, it is important to know how the design of interfaces affects users' perception of AI-powered systems for healthcare. This paper examines the influence of explainable artificial intelligence (XAI) on user trust, usability, and confidence in a healthcare decision support interface. We implemented a prototype of the diabetes risk prediction based in machine learning and work with two interface versions: black-box, which displays only the prognostic output information; explainable, which also returns human understandable justifications for those predictions. A controlled user study was performed to compare user reactions between the two interfaces with personalized usability and trust measures. Experimental results demonstrate that the explainable interface has a clear and significant improvement on perceived trust, usability, and user confidence over the black-box one. Our work demonstrates the significance of interpretability in healthcare AI systems, and offers actionable design recommendations on developing accountable and user-centered AI-enabled applications for health concerns. This paper offers evidence for the emerging area of human–AI interactions in healthcare.**

***Index Terms*—Human–Computer Interaction, Explainable AI, Human-Centered AI, Healthcare Applications, User Trust.**

I. Introduction

Applications of artificial intelligence (AI) in healthcare such as disease prediction, clinical decision-making and risk assessment have been rapidly developed. The recent development with machine learning is that AIs can achieve impressive predictive performance across many medical tasks. Yet, except these technical progresses, the implementation of AI system in healthcare environments is still underutilized. A key impediment to usage is the opacity of AI decision-making that results in low trust and reluctancy among clinical workers and end users [1], [2].

From an HCI point of view, trustworthiness, usability and user confidence are important drivers of human behaviours towards AI-driven healthcare systems. Existing studies emphasize that even if AI systems exhibit high accuracy of prediction, they are not infrequently rejected due to their lack of understanding for the way a user perceives and interacts with the system [3]. In healthcare, where the stakes are high and decisions can have critical implications, clinicians are especially sensitive to opaque/balck-box AI behavior.

Explainable Artificial Intelligence has emerged as a way to tackle these issues by explicating AI predictions in a manner that humans can comprehend. Explanation of model's output through XAI, seeks to achieve greater transparency, trustworthiness and promotion of informed decision making. Some recent studies indicate that such an explainable interface can have a positive effect on the user's mental model of AI, including perceived reliability, in high-stakes healthcare scenarios [4], [5].

Despite increasing interest in XAI, empirical work presents conflicting results on the efficacy of enhancing trust and usability. A few experiments do show that confidence and trust are enhanced by the provision of explanations, but others suggest that the wrong sort of explanation can overload a user or even reduce trust [6]. This inconsistency indicates that the influence of explainability is a function not only of explanation presence/absence, but also how explanations are surfaced via the user interface. Therefore, it is an open research question in HCI and healthcare AI communities on how explainability and interface design interact with each other.

Also, much of the existing work has concentrated on algorithmic approaches for explainability more than evaluating explainable interfaces from an HCI point of view in empirical user studies [7]. Controlled experiential studies are needed to systematically compare between black-box and explainable interfaces while measuring critical user-centered outcomes such as trust, usability, and confidence.

To bridge this gap, a human-centric assessment of explainable AI is performed in healthcare decision support. A prototype diabetes risk prediction system is designed whose interfaces differ in transparency: black-box (prediction-only) and explainable (with explanation). A controlled user study is conducted to examine trust and perceived usability in the latter type of interface designs that persuade AIs, confidence in AIs. Contributions of this work: it describes (1) the design and realization of an explainable decision support prototype for AI-based healthcare decision supports with XAI/NEXAI interfaces, (2) empirical investigation into user perception via a set of structured user studies; and (3) key takeaways valuable for designing reliable human-acceptable AI systems

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in healthcare.

II. Literature Review

Explainable Artificial Intelligence (XAI) has drawn great attention in the application of healthcare for AI system use in clinical decision-making, disease prediction and risk analysis. Early work highlighted the importance of trustworthiness, transparency, and access as key for successful deployment of AI in medicine (in particular high-stakes scenarios) [1]. In addition, the interpretability allows for improving bridging high-prediction performance to effective clinical deployment which is justified by recent works [8]–[10]. These findings indicate that opaque AI systems can compromise user trust and subsequent adoption into clinical practice.

From a human–computer interaction perspective, explainability is now largely seen as a human–computer interaction issue rather than solely as a technical property of AI models. Systematic reviews that investigate the incorporation of HCI with XAI in the health domain, emphasise that most current systems are often developed with little concern for user perception or interaction, as well as cognition [2]. Human–AI interaction studies also reveal that user experience, cognitive load, and formation of mental model are mediating the motivation to use and trust towards a system [11]. User-centered studies also show that the effectiveness of explanation depends on different aspects including user expertise, task context and interface presentation [12], [13]..

A number of frameworks have been suggested for assisting the development of human-centric and trustworthy AI systems in healthcare. De Silva et al. proposed a hierarchical approach, accounting for explainability, trust and usability at various levels of AI system design [3]. Supplement reviews on Clinical Decision Support Systems to note explanation methods should be adapted with clinical workflows and in-line with real-world decision-making practices [14], [15]. Although such frameworks provide useful theoretical sensitizing, many are based on secondary analysis and retrospective case study with relatively little validation through controlled experimental user studies.

Trust has been habitually cited as one of the most important contributors for user adoption toward AI-based healthcare technology. Feedback and empirical evidence indicate that explainability would increase trust depending on the explanation: clarity of contents, relevance to user expectation, manner of presentation [6]. Comparative researches show that, in practice, clinicians favor interpretable stress explanations which are actionable and context-aware than being technical oriented [16]. Furthermore, meta-analytic results also suggest that the association between explainability and trust is multifaceted and nonlinear, emphasizing the role of usability in mediating user responses [17], [18].

Explainable user interfaces as a crucial aspect of human-centered AI systems are recently studied more and more. Studies that survey explainable interfaces in healthcare demonstrate the absence of common protocols for assessing user experience (e.g., trust, usability, and confidence) [7]. User-centred evaluation studies also find that many XAI systems have a focus on algorithmic transparency but lack appropriate emphasis on how explanations are provided via the GUI [4], [19]. Standalone studies such as those addressing the older generations further emphasize that explanation design should be tailored to the capabilities and expectations of different groups [20].

Previous work has achieved remarkable advances in XAI approaches, trust modeling and human-in-the-loop AI systems. There are however little experimental evidences why such additional explainability will indeed improve HCI-related outcomes. Specifically, more evidence is required to elucidate the extent to which interface level explainability influences trust, perceived usability and user confidence in health decision systems. This work fills this gap by performing a controlled user study comparing black-box and explainable interfaces in the healthcare context, adding to the emerging human-centered XAI literature.

III. Methodology

Figure 1 depicts the proposed methodology to assess impact of Explainable AI (XAI) in clinical decision support in healthcare. The workflow comprises four main stages: 1) data collection and preparation, 2) model conception and development, 3) front-end design planning and execution, and finally 4) empirical user evaluation.

A. Dataset Description

The DiaBD (Diabetes Bangladesh) dataset was employed to conduct this study. In addition, the dataset contains important demographic and clinical features such as age, gender, BMI index, blood pressure and so on; physical activity level; family history; glucose measurements and a particular variable to identify each record among others with its corresponding diabetes status. Simple pre-processing with normalization and categorical encoding was performed to prepare the data for model training and testing.

TABLE I
Description of Dataset Attributes

Attribute Name	**Description**	**Data Type**
Age	Age of the participant (years)	Numerical
Gender	Biological sex of the participant	Categorical
BMI	Body Mass Index	Numerical
BloodPressure	Resting blood pressure level	Numerical
Glucose	Blood glucose level	Numerical
PhysicalActivity	Daily physical activity level	Categorical
FamilyHistory	Family history of diabetes	Binary
SmokingStatus	Smoking habit of participant	Categorical
DietType	Type of daily diet	Categorical
Diabetes	Diabetes diagnosis label	Binary

Table I shows attributes of the DiaBD dataset with their data types and brief descriptions when they are used in this study. Features comprise demographic, clinical and lifestyle-related characteristics which are considered for diabetes risk assessment. These characteristics act as inputs to the AI model to predict the outcomes and make interpretable insights.

B. Dataset Preparation

First, to simulate real-world healthcare environments we created a dataset of 200 synthetic patients with clinically relevant characteristics (e.g., age, body mass index, blood pressure, cholesterol levels and glucose level). Data preprocessing consisted in missing values treatment, continuous variables normalization and categorical features encoding, for compatibility with downstream machine learning model. Variable selection methods were used to select the most predictive predictors, improving model performance and interpretability.

C. Model Development

In the model development phase, several supervised machine learning models (i.e., Support Vector Classifier, Logistic Regression and Decision Tree) were considered for the prediction of diabetes risk. The model with the best validation set performance in terms of F1-score and predictive accuracy was chosen. The explainability module was incorporated into the model using post-hoc XAI methods, thereby delivering human-interpretable explanations for individual predictions and keeping users informed on how each feature affected the model's decision.

D. Interface Design

The interface design resulted in two prototypes: a black-box interface that only presents the predictions, and an explainable one showing predictions as well as explanation on feature level. The black-box interface was designed to provide uninformative information display, while the explainable interface shows relative contribution of input features through duality between visual and textual explanations. In particular post-hoc feature attribution explanations using SHAP (SHapley Additive exPlanations) were leveraged to measure and expose the influence of each feature on the model's prediction improving transparency and reproducibility of the system.

Human-centered principles were the basis of the design of interface elements (e.g., probability scores, bar charts and succinct explanations) to ensure clearness, interpretability and trust [3], [4]. Visual hierarchy and explanation position were taken into account to reduce cognitive load. Pilot tests were used to improve layout uniformity, visibility and usability.

E. Empirical User Evaluation

The empirical user evaluation phase used a between-subject controlled experiment where participants worked with both types of interface. Perceived usability, trust, and confidence were measured with previously validated instruments (System Usability Scale (SUS), modified trust questionnaire). Performance on task-based measures (accuracy of trust calibration in interpreting AI predictions, decision time) were also logged.

Statistical analysis Whether participants learned more with the explainable interface than with the black-box interface was examined by descriptive statistics, paired- samples t-tests and ANOVA for quantifying trends on effects of interface conditions and qualitative feedback to provide a deeper user reaction perspective and experience.

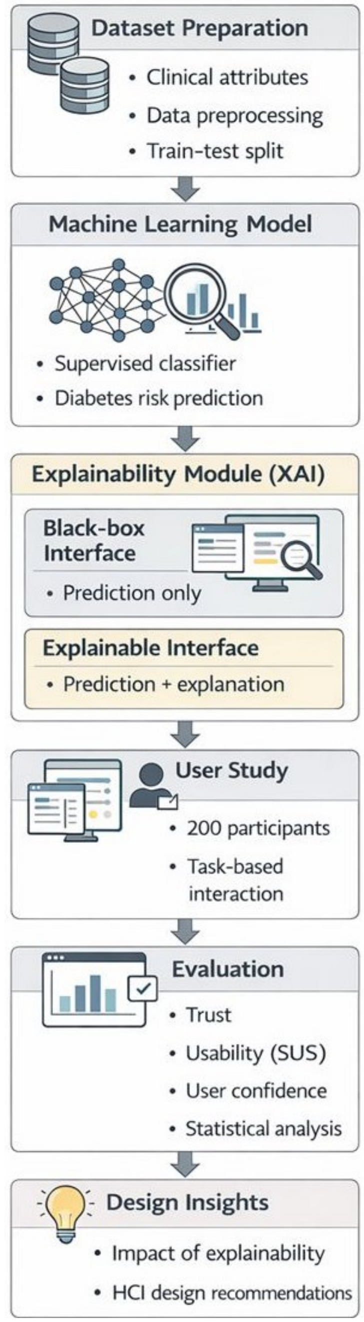

Fig. 1. Wokflow of the Proposed HCI based XAI Healthcare System

This approach guarantees a thorough investigation of the relationship of model explainability, interface design, and user perception as well as for designing trustworthy AI systems in healthcare. Figure 1 Visual representation of the workflow, depicting stepwise combination of data preparation, predictive modeling, explainable interface design and rigorous user evaluation.

IV. Results and Discussions

This section presents and contrasts the performance of different machine learning algorithms, and trust, usability and confidence scores from a user perspective. The integrated dataset was then randomly partitioned into 80% of the training data and the remaining 20% for testing or validation.

TABLE II
COMPARISON OF ML MODELS

Algorithm	Accuracy-(%)	Precision-(%)	Recall-(%)	F1 Score(%)
SVM	93.57	93.35	92.46	92.54
Logistic Regression	93.47	93.23	92.34	92.29
Decision Tree	90.59	90.85	90.52	90.56

A comparison of selected machine learning models is given by table II. Among them, SVM achieves the best overall performance, attaining the highest accuracy, precision, recall, and F1-score. Logistic Regression demonstrates comparable results, performing slightly lower than SVM across all metrics. In contrast, the Decision Tree model shows comparatively lower performance, though it still maintains reasonably high accuracy and balanced precision–recall values. Overall, the results indicate that SVM and Logistic Regression are more effective for this task than the Decision Tree model.

The confusion matrices in Fig. 2 show that the Support Vector Classifier (SVC) and Logistic Regression models perform similarly, both achieving strong results in identifying negative cases but struggling somewhat with positive ones. The SVC correctly classifies 861 negatives and 57 positives, with 19 false negatives and 121 false positives, while Logistic Regression catches slightly more positives (58) but at the cost of more false positives (156).

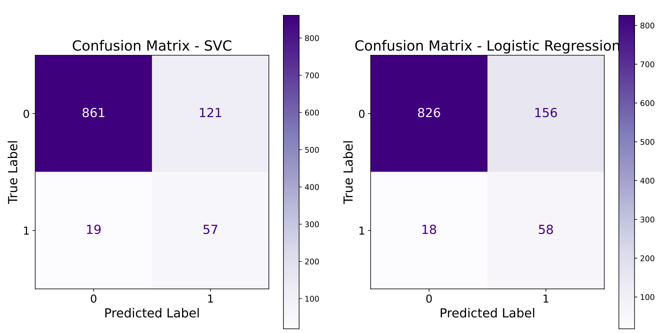

Fig. 2. Confusion matrix of SVM and Logistic Regression

In contrast, the Decision Tree model's confusion matrix in Fig. 3 is more conservative: it achieves the highest true negatives (862) but misses a larger portion of positive cases, with 34 false negatives and only 42 true positives. Overall, SVC and Logistic Regression balance sensitivity and specificity better, whereas the Decision Tree favors specificity at the expense of recall. This behavior indicates that the Decision Tree is biased toward the majority class, making it less suitable for detecting minority-class instances in imbalanced datasets compared to SVC and Logistic Regression.

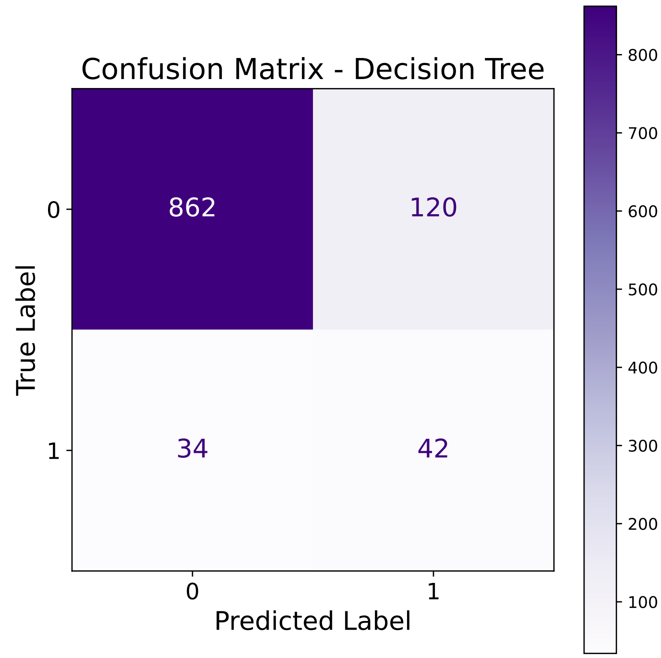

Fig. 3. Confusion matrix of Decision Tree

The ROC-AUC shown in Fig. 4 curve compares the classification performance of the three models. Both the Support Vector Classifier (AUC = 0.881) and Logistic Regression (AUC = 0.879) achieve high scores, indicating strong discriminative ability and nearly identical performance in distinguishing between classes. In contrast, the Decision Tree model performs noticeably worse, with an AUC of 0.702, reflecting weaker separation between positive and negative cases. All in all, the curve indicates that SVC and Logistic Regression are more reliable for this specific task, whereas the Decision Tree has limited predictive power.

The precision–recall shown in Fig. 5 compares the performance of the three classifiers. SVC performs the best with an average precision of 0.920, very closely followed by Logistic Regression at 0.890 and Decision Tree having about 0.780 as the score/score. SVC and LR both achieve higher precision levels all over the recall spectrum, based mainly on better reliability to predict positive cases in decision-making rather than a particular better performance under class imbalance situation. As a whole, these results suggest that SVC and Logistic Regression better predict performance, but the Decision Tree somewhat lags behind.

The box plot as shown in Fig. 6 for trust ratings on "Blackbox" and "Explainable" user interface. In the case of the Blackbox UI, the median trust score is around 3.0, with an interquartile range of nearly from 2.5 to 3.5, and drawing whiskers very near 2.0 and 4.0 which shows lower and more variable user trust. In contrast, the Explainable UI displays a median trust of around 4.5 and with a tighter interquartile range between 4.0-4.80 and whiskers stretching up to 5.0 indicating both stronger and more constant trust among users. These results unambiguously show that interface designs that

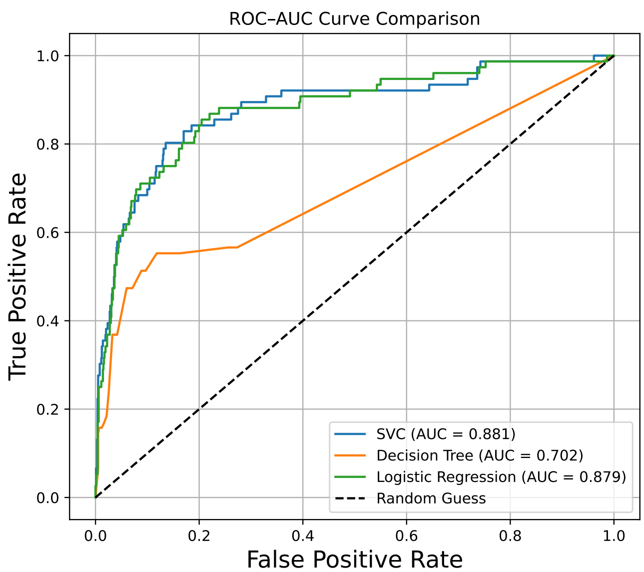

Fig. 4. ROC-AUC Curve of Used Models

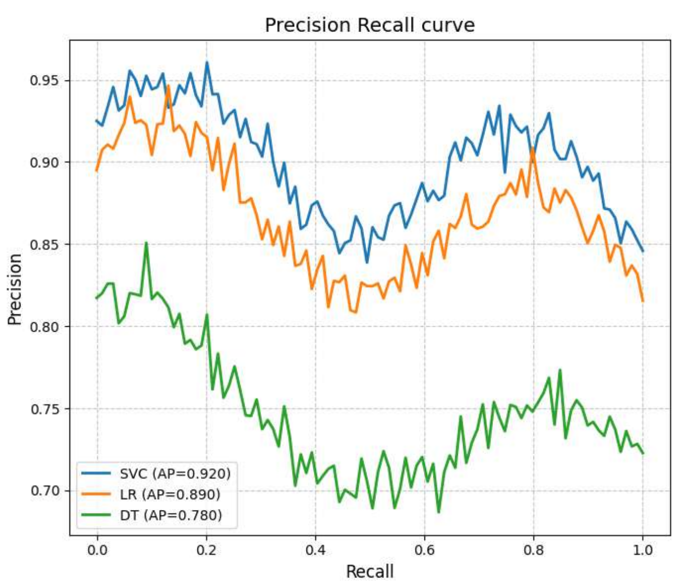

Fig. 5. Precision-Recall Curve of Used Models

are explainable promote much stronger and predictable levels of user trust than blackbox interfaces.

The box plot in Fig. 7 show the SUS Usability results for both interfaces. In contrast, the median is lower ($\sim 65-70$ m) and terrain more spread out ($\sim 55-75$) which suggests low usability and wide variability amongst users. That's closer to something like the explainable UI with a higher median ($\sim$ 80–82) but also lower range ($\sim$ 76–86), i.e., easier to use and less inconsistent. The low range of scores shows a consistency in agreement among users, and higher SUS values imply better learnability, clarity and satisfaction in general. The results those are found suggest that the interpretable interfaces are perceived to be more intuitive and trustful than the black-box designs.

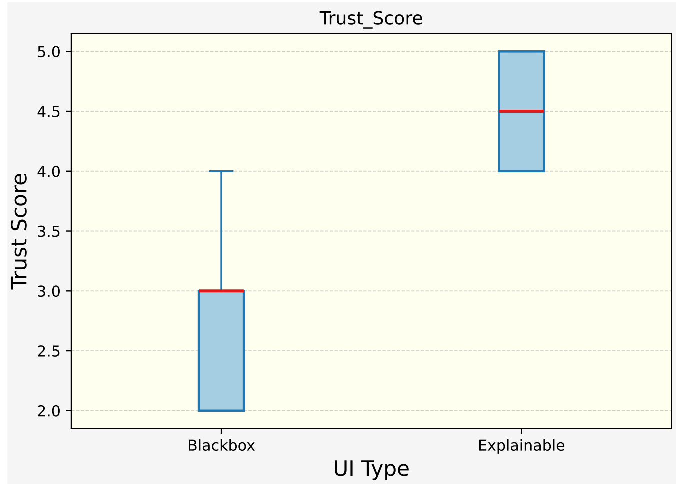

Fig. 6. Blackbox UI vs XAI UI Trust Score Comparison

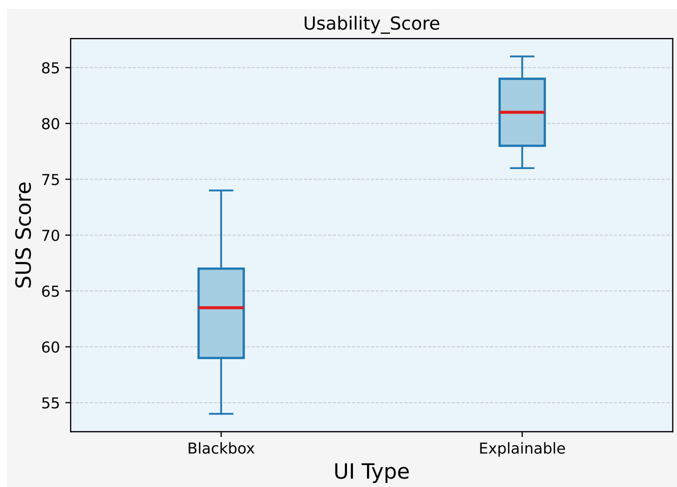

Fig. 7. Blackbox UI vs XAI UI SUS Score Comparison

The box plot in Fig. 8 shows clear differences in confidence scores between the two interfaces. The Blackbox UI has a lower median around 3.0 with a wider spread, indicating lower and less consistent user confidence. In contrast, the Explainable UI achieves a higher median near 4.5, with most values concentrated between 4.0 and 5.0, reflecting stronger and more consistent confidence. The smaller spread in the Explainable condition suggests greater agreement among participants, while the wider variation in the Blackbox condition indicates uncertainty and mixed perceptions. Statistical analysis also shows a significant difference ($p < 0.05$), confirming that the improvement is meaningful. Overall, explainable interfaces enhance user confidence, trust, and reliable user decisions compared to black-box designs.

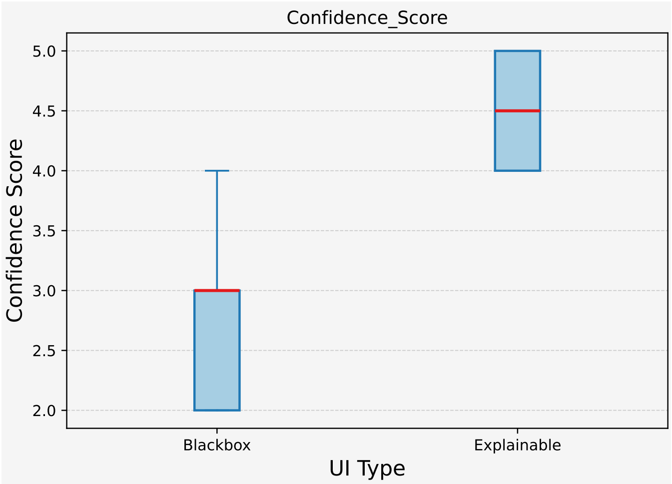

Fig. 8. Blackbox UI vs XAI UI Confidence Score Comparison

A. Error Analysis

Although the overall results were relatively good, there were some misclassified instances. The majority of the errors were made on these borderline user responses and overlappings. As an example, some Blackbox UI ratings were not consistent with a fixed confidence or usability assessments. These mistakes reveal the difficulty of understanding subjective measures, and indicate the need for a larger sample with more diverse subjects in future research to eliminate bias and enhance resilience.

V. Conclusion

The research provided a human-focused analysis on eXplainable AI (XAI) for healthcare support by determining how explainability influences trust, usability and confidence. Both the SVM and Logistic Regression models yielded strong predictive performance at 93% accuracy, with precision and recall being well-balanced. The quantitative user study results indicated that the Explainable UI resulted in a higher median of trust (~ 4.5) vs. (~ 3.0), SUS usability score ($\sim 80 - 82$) vs. ($\sim$ $65 - 70$) and more confidence ($\sim$ 4.5 vs. $\sim$ 3.0) than the Blackbox UI, with less spread across participants as well. These results demonstrate that well-designed explainable interfaces can significantly improve user's perception and interaction quality. The generalizability is limited to an in-laboratory experiment using synthetic data, larger studies including real patients and clinical data as well as external validation are the subject of future work.

The generalization could be further enhanced by including more complex real clinical datasets in the future. Investigating advanced explainability approaches, adaptive interfaces for varying expertise levels, and longitudinal clinician studies may also help to enhance trust and promote the real-world use of explanation approaches in health care HCI.

References

[1] C. M. Cutillo, K. R. Sharma, L. Foschini, S. Kundu, M. Mackintosh, K. D. Mandl, and M. in Healthcare Workshop Working Group Beck Tyler 1 Collier Elaine 1 Colvis Christine 1 Gersing Kenneth 1 Gordon Valery 1 Jensen Roxanne 8 Shabestari Behrouz 9 Southall Noel 1, "Machine intelligence in healthcare—perspectives on trustworthiness, explainability, usability, and transparency," *NPJ digital medicine*, vol. 3, no. 1, p. 47, 2020.

[2] M. Nazar, M. M. Alam, E. Yafi, and M. M. Su'ud, "A systematic review of human–computer interaction and explainable artificial intelligence in healthcare with artificial intelligence techniques," *IEEE Access*, vol. 9, pp. 153316–153348, 2021.

[3] C. De Silva, T. Halloluwa, and D. Vyas, "A multi-layered research framework for human-centered ai: Defining the path to explainability and trust," *arXiv preprint arXiv:2504.13926*, 2025.

[4] I. Donoso-Guzmán, K. S. Kacafírková, M. Szymanski, A. Jacobs, D. Parra, and K. Verbert, "A systematic review of user-centred evaluation of explainable ai in healthcare," *arXiv preprint arXiv:2506.13904*, 2025.

[5] A. Johannssen and N. Chukhrova, "The crucial role of explainable artificial intelligence (xai) in improving health care management," *Health Care Management Science*, pp. 1–6, 2025.

[6] R. Rosenbacke, Å. Melhus, M. McKee, and D. Stuckler, "How explainable artificial intelligence can increase or decrease clinicians' trust in ai applications in health care: systematic review," *Jmir Ai*, vol. 3, p. e53207, 2024.

[7] J. Cálem, C. Moreira, and J. Jorge, "Intelligent systems in healthcare: A systematic survey of explainable user interfaces," *Computers in Biology and Medicine*, vol. 180, p. 108908, 2024.

[8] Z. Sadeghi, R. Alizadehsani, M. A. Cifci, S. Kausar, R. Rehman, P. Mahanta, P. K. Bora, A. Almasri, R. S. Alkhawaldeh, S. Hussain, *et al.*, "A review of explainable artificial intelligence in healthcare," *Computers and Electrical Engineering*, vol. 118, p. 109370, 2024.

[9] V. Chamola, V. Hassija, A. R. Sulthana, D. Ghosh, D. Dhingra, and B. Sikdar, "A review of trustworthy and explainable artificial intelligence (xai)," *IEEe Access*, vol. 11, pp. 78994–79015, 2023.

[10] A. A. Adeniran, A. P. Onebunne, and P. William, "Explainable ai (xai) in healthcare: Enhancing trust and transparency in critical decision-making," *World Journal of Advanced Research and Reviews*, vol. 23, no. 3, pp. 2647–2658, 2024.

[11] I. Shanthi, A.-N. Seow, and J.-J. Chang, "Understanding human-ai interaction in healthcare: The mediating role of trust and moderating influence of cognitive load," *Malaysia Journal of Invention and Innovation*, 2024.

[12] Y. Rong, T. Leemann, T.-T. Nguyen, L. Fiedler, P. Qian, V. Unhelkar, T. Seidel, G. Kasneci, and E. Kasneci, "Towards human-centered explainable ai: A survey of user studies for model explanations," *IEEE transactions on pattern analysis and machine intelligence*, vol. 46, no. 4, pp. 2104–2122, 2023.

[13] F. A. Cadelina, "Advanced ai methodologies for enhancing user experience in human-computer interaction," 2024.

[14] M. Salimparsa, K. Sedig, D. J. Lizotte, S. S. Abdullah, N. Chalabianloo, and F. T. Muanda, "Explainable ai for clinical decision support systems: Literature review, key gaps, and research synthesis," in *Informatics*, vol. 12, p. 119, MDPI, 2025.

[15] Y. Liu, C. Liu, J. Zheng, C. Xu, and D. Wang, "Improving explainability and integrability of medical ai to promote health care professional acceptance and use: mixed systematic review," *Journal of Medical Internet Research*, vol. 27, p. e73374, 2025.

[16] L. Bergomi, G. Nicora, M. A. Orlowska, C. Podrecca, R. Bellazzi, C. Fregosi, F. Salinaro, M. Bonzano, G. Crescenzi, F. Speciale, *et al.*, "Which explanations do clinicians prefer? a comparative evaluation of xai understandability and actionability in predicting the need for hospitalization," *BMC Medical Informatics and Decision Making*, vol. 25, no. 1, p. 269, 2025.

[17] Z. Atf and P. R. Lewis, "Is trust correlated with explainability in ai? a meta-analysis," *IEEE Transactions on Technology and Society*, 2025.

[18] E. Zeng, H. Liu, L. Wang, W. Zhao, and Y. Feng, "Literature review: A study of xai user experience in healthcare: Transparency and doctor-patient trust construction based on ai-assisted diagnosis," *Frontiers in Interdisciplinary Applied Science*, vol. 2, no. 1, pp. 21–33, 2025.

[19] A. Alaqsam and C. Sas, "Systematic review of xai tools for ai-hci research," *Electronic Workshops in Computing*, 2024.

[20] X. Huang, Z. Zhang, F. Guo, X. Wang, K. Chi, and K. Wu, "Research on older adults' interaction with e-health interface based on explainable artificial intelligence," *ArXiv*, vol. abs/2402.07915, 2024.